

```
In [21]: n = int(input("Enter n value :"))
        for i in range(1,n+1):
            for j in range(1,i+1):
                print('*',end='')
            print('')
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [12]: n = int(input("Enter n value :"))
        for i in range(1,n+1):
            for j in range(1,n-i+1):
                print('*',end='')
            print('')
```

```
* * * * *
* * * *
* * *
* *
*
```

```
In [7]: n = int(input("Enter n :"))
        for i in range(n):
            for j in range(n):
                print("*",end=" ")
            print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [16]: n = int(input("Enter n value :"))
        for i in range(n+1):
            for j in range(0,n-i):
                print(' *',end='')
            print('')
```

```
* * * * *
* * * *
* * *
* *
*
```

```
In [20]: n = int(input("Enter n :"))
        for i in range(1,n+1):
            for j in range(1,i+1):
                print(j,end=' ')
            print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [34]: n = int(input("Enter n :"))
for i in range(1,n+1):
    for j in range(1,n+1):
        print(' ',end=' ')
    for k in range(1,k+1):
        print('* ',end=' ')
    print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [2]: s = "  fly me  to  the moon  "

list1 = s.split()
print(list1)
```

```
['fly', 'me', 'to', 'the', 'moon']
```

```
In [5]: print(1 and 3 and 0)
```

```
0
```

```
In [4]: g = [1,2,3]
s = [1,1]
g.sort()
s.sort()
j = 0
satisfy = 0

for i in range(min(len(g),len(s))):
    if g[i] == s[j]:
        satisfy += 1
        j += 1
    else:
        continue
print(satisfy)
```

```
1
```

```
In [1]: # largest element in an array:
array = [10,9,3,12,14,31]

largest = array[0]

for i in array:
    if largest <= i:
        largest = i
print(largest)
```

```
31
```

```
In [2]: #smallest element in an array

array = [10,9,3,12,14,31]

smallest = array[0]

for i in array:
```

```

    if smallest >= i:
        smallest = i
print(smallest)

```

3

In [8]: *#second largest element in an array*

```

array = [0,1]

largest = array[0]
second_largest = -1
for i in array:
    if largest <= i:
        largest = i
for i in array:
    if i > second_largest and i != largest:
        second_largest = i
print(second_largest)

```

0

In [12]: *#second smallest element in an array*

```

array = [10,9,3,12,14,31]

smallest = array[0]
second_smallest = -1
for i in array:
    if smallest >= i:
        smallest = i
print(smallest)
for i in array:
    if i > second_smallest and i == smallest:
        second_smallest = i
print(second_smallest)

```

3

3

In [32]:

```

array = [10,1]
largest = array[0]
s1 = -1
print(len(array))
for i in range(1,len(array)):
    if array[i] > largest:
        s1 = largest
        largest = array[i]
    elif array[i] < largest and array[i] > s1:
        s1 = array[i]
print(s1)

```

2

1

In [45]: *#check wether the array is sorted or not*

```

#array = [10,9,3,12,14,31]
array = [1,2,3,4,5,6,7,8,9]
def sortedarr():
    for i in range(1,len(array)):
        if array[i] >= array[i-1]:
            first = array[i]
        else:

```

```

        return False
    return True
sortedarr()

```

Out[45]: True

```

In [8]: n = int(input("Enter n value :"))
        for i in range(n+1):
            for j in range(i):
                print('*',end=' ')
            print()

```

```

*
* *
* * *
* * * *
* * * * *

```

```

In [9]: for i in range(5):
        for j in range(5-i):
            print('*',end=' ')
            j -= 1
        print()

```

```

* * * * *
* * * *
* * *
* *
*

```

```

In [7]: ##### for i in range(5):
        for j in range(5-i):
            print('*',end=' ')
            j -= 1
        print()
        for i in range(2,5+1):
            for j in range(i):
                print('*',end=' ')
            print()

```

```

* * * * *
* * * *
* * *
* *
*
* *
* * *
* * * *
* * * * *

```

```

In [33]: n = int(input("Enter n value :"))
        for i in range(1,n+1):
            for j in range(i):
                print(i,end='')
            print()

```

```
1
22
333
4444
55555
```

```
In [32]: for i in range(1,5+1):
          for j in range(1,i+1):
              print(j,end=' ')
          print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [47]: n = int(input("Enter n value :"))

          for i in range(1,n+1):
              for j in range(1,n-i+2):
                  print(j,end=' ')

              print()
```

```
12345
1234
123
12
1
```

```
In [ ]:
```

```
In [6]: from collections import Counter
          word = "xyzxyzxyzxyz"
          frequency = Counter(word)
          print(sum(frequency.values()))
```

```
12
```

```
In [1]: n = int(input("Enter n value :"))

          for i in range(1,n+1):
              for j in range(1,n-i+2):
                  print(j,end=' ')
              print()
          for i in range(2,n+1):
              for j in range(1,i+1):
                  print(j,end=' ')
              print()
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [15]: n = int(input("Enter n value :"))

for i in range(1,n*2):
    for j in range(1,n-i+2):
        print(j,end=' ')
    print()
if i>= 5 and j == 5:
    for j in range(1,i+1):
        print(j,end=' ')
    print()
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
In [16]: n = int(input("Enter n value: "))

for i in range(1, n * 2):

    cols = (n - i + 1) if i <= n else (i - n + 1)

    for j in range(1, cols + 1):
        print(j, end=' ')
    print()
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [42]: n = int(input("Enter n value :"))

for i in range(n):
    # print space
    for j in range(n-i-1):
        print(' ',end=' ')
    # print star
    for k in range(2*i+1):
        print('*',end=' ')
    # print space
    for m in range(n-i-1):
        print(' ',end=' ')
    print()
```

```

      *
    * * *
  * * * * *
* * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *

```

```
In [43]: n = int(input("Enter n value :"))
```

```
for i in range(n):  
    # print space  
    for j in range(i):  
        print(' ',end=' ')  
    # print star  
    for k in range(2*n-(2*i +1)):  
        print('*',end=' ')  
    # print space  
    for m in range(i):  
        print(' ',end=' ')  
    print()
```

```

* * * * * * * * * * * * * * * * *
*
* * * * * * * * * * * * * * * *
*
* * * * * * * * * * * * * * *
*
* * * * * * * * * * * * *
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*
* * * * * * * * *
*
* * * * * * *
*
* * * * *
*
* * *
*
*
```

```
In [5]: a = 1&2
        print(a)
```

0

```
In [2]: n = int(input("Enter n value :"))
```

```
for i in range(1, 2*n):
    if i <= n:
        stars = i
    else:
        stars = 2 * n - i
    for j in range(stars):
        print('*', end='')
    print()
```

```

*
**
***
****
*****
*****
****
***
**
*

```

In [4]: `print((1.0 + 1.0) == 2.0)`

True

In [2]: `# Pattern 1`
`# 11111`
`# 11111`
`# 11111`
`# 11111`
`# 11111`
`# 11111`

`n = int(input("Enter n value :"))`
`for i in range(1,n+1):`
 `for j in range(1,n+1):`
 `print("1",end="")`
 `print()`

11111
 11111
 11111
 11111
 11111
 11111

In [3]: `# Pattern 2`
`# *****`
`# *****`
`# *****`
`# *****`
`# *****`

`n = int(input("Enter n value :"))`
`for i in range(1,n+1):`
 `for j in range(1,n+1):`
 `print("*",end="")`
 `print()`

In [4]: `# Pattern 3`
`# AAAAA`
`# AAAAA`
`# AAAAA`
`# AAAAA`
`# AAAAA`

`n = int(input("Enter n value :"))`

`for i in range(n):`
 `for j in range(n):`
 `print("A",end='')`
 `print()`

AAAAA
 AAAAA
 AAAAA
 AAAAA
 AAAAA

```
In [5]: # Pattern 4
# 11111
# 22222
# 33333
# 44444
# 55555
n = int(input("Enter n value :"))

for i in range(1,n+1):
    for j in range(1,n+1):
        print(i,end='')
    print()
```

11111
 22222
 33333
 44444
 55555

```
In [13]: # Pattern 5
# AAAAA
# BBBBB
# CCCCC
# DDDDD
# EEEEE
n= int(input("Enter n value :"))

for i in range(1,n+1):
    for j in range(1,5+1):
        print(chr(64+i),end='')
    print()
```

AAAAA
 BBBBB
 CCCCC
 DDDDD
 EEEEE
 FFFFF
 GGGGG
 HHHHH
 IIIII
 JJJJJ

```
In [14]: # Pattern 6
# 12345
# 12345
# 12345
# 12345
# 12345

n = int(input("Enter n value :"))

for i in range(1,n+1):
    for j in range(1,n+1):
```

```
    print(j,end='')
    print()
```

```
12345
12345
12345
12345
12345
```

```
In [16]: # Pattern 7
# ABCDE
# ABCDE
# ABCDE
# ABCDE
# ABCDE

n = int(input("Enter n value :"));
for i in range(1,n+1):
    for j in range(1,n+1):
        print(chr(64+j),end='')
    print()
```

```
ABCDE
ABCDE
ABCDE
ABCDE
ABCDE
```

```
In [21]: # Pattern 8
# 54321
# 54321
# 54321
# 54321
# 54321

n = int(input("Enter n value :"))

for i in range(1,n+1):
    for j in range(n,0,-1):
        print(j,end='')
    print()
```

```
54321
54321
54321
54321
54321
```

```
In [31]: # Pattern 9
# EDCBA
# EDCBA
# EDCBA
# EDCBA
# EDCBA

n = int(input("Enter n value :"))

for i in range(n):
    for j in range(ord('A')+n-1,64,-1):
```

```
        print(chr(j),end='')
    print()
```

EDCBA
EDCBA
EDCBA
EDCBA
EDCBA

```
In [49]: # Pattern 10
# 55555
# 44444
# 33333
# 22222
# 11111

n = int(input("Enter n value :"))

for i in range(n,0,-1):
    for j in range(1,n+1):
        print(i,end='')
    print()
```

55555
44444
33333
22222
11111

```
In [71]: # Pattern 11
# A B A B A
# B A B A B
# A B A B A
# B A B A B
# A B A B A

n = int(input("Enter n value :"))

for i in range(1,n+1):
    for j in range(1,n+1):
        if (j+i)%2==0:
            print('A',end=' ')
        else:
            print('B',end=' ')
    print()
```

A B A B A
B A B A B
A B A B A
B A B A B
A B A B A

```
In [70]: # Pattern 12
# 0 1 0 1 0
# 1 0 1 0 1
# 0 1 0 1 0
# 1 0 1 0 1
# 0 1 0 1 0

n = int(input("Enter n value :"))
for i in range(1,n+1):
```

```

for j in range(1,n+1):
    print((i+j)%2,end=' ')
print()

```

```

0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0

```

```

In [83]: # Pattern 13
# 1 2 3 4 5
# 2 3 4 5 6
# 3 4 5 6 7
# 4 5 6 7 8
# 5 6 7 8 9

n = int(input("Enter n values :"))

for i in range(1,n+1):
    for j in range(i,n+i):
        print(j,end=' ')
    print()

```

```

1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9

```

```

In [90]: # Pattern 14
# 1 2 3 4 5
# 6 7 8 9 10
# 11 12 13 14 15
# 16 17 18 19 20
# 21 22 23 24 25

n = int(input("Enter n value :"))
for i in range(1,(n*n)+1,5):
    for j in range(i,n+i):
        print(j,end=' ')
    print()

```

```

1 2 3 4 5
6 7 8 9 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25

```

```

In [113... # Pattern 15
# 2 4 6 8 10
# 12 14 16 18 20
# 22 24 26 28 30
# 32 34 36 38 40
# 42 44 46 48 50

# n = int(input("Enter n vsLue :"))

# for i in range(2,(n*10)+1,10):
#     for j in range(i,(n*2)+1,2):

```

```
#         print(j,end=' ')
#     print()

n = int(input("Enter n value :"))
for i in range(2,n*10,10):
    for j in range(i,10+i,2):
        print(j,end=' ')
    print()
```

```
2 4 6 8 10
12 14 16 18 20
22 24 26 28 30
32 34 36 38 40
42 44 46 48 50
```

In [137...

```
# Pattern 16
# A B C D E
# F G H I J
# K L M N O
# P Q R S T
# U V W X Y

n = int(input("Enter n value :"))

for i in range(65,90,5):
    for j in range(i,i+5):
        print(chr(j),end=' ')
    print()
```

```
A B C D E
F G H I J
K L M N O
P Q R S T
U V W X Y
```

In [145...

```
# Pattern 17
# 1 2 3 4 5
# A B C D E
# 1 2 3 4 5
# A B C D E
# 1 2 3 4 5

def print_nums(n):
    for i in range(1,n+1):
        print(i,end=' ')
    print()

def print_letters(n):
    for i in range(65,65+n):
        print(chr(i),end=' ')
    print()

n = int(input("Enter n value : "))

for i in range(round(n/2)+1):
    print_nums(n)
    print_letters(n)
```

```

1 2 3 4 5
A B C D E
1 2 3 4 5
A B C D E
1 2 3 4 5
A B C D E

```

In [156...

```

# Pattern 18
# 1 1 1 1 1
# B B B B B
# 3 3 3 3 3
# D D D D D
# 5 5 5 5 5

def numbers_print(num,n):
    for i in range(n):
        print(num,end=' ')
    print()
def letter_print(letter,n):
    for i in range(n):
        print(chr(letter),end=' ')
    print()
n = int(input("Enter n value :"))
for i in range(1,n+1):
    if i%2==1:
        numbers_print(i,n)
    else:
        letter_print(65+i-1,n)

```

```

1 1 1 1 1
B B B B B
3 3 3 3 3
D D D D D
5 5 5 5 5

```

In [169...

```

# Pattern 19
# 5 5 5 5 5
# 7 7 7 7 7
# 9 9 9 9 9
# 11 11 11 11 11
# 13 13 13 13 13

n = int(input("Enter n value :"))

for i in range(n,n*3,2):
    for j in range(n,n*3,2):
        print(i,end=' ')
    print()

```

```

5 5 5 5 5
7 7 7 7 7
9 9 9 9 9
11 11 11 11 11
13 13 13 13 13

```

In [181...

```

# Pattern 20
# Z Z Z Z Z
# Y Y Y Y Y
# X X X X X
# W W W W W
# V V V V V

```

```

n = int(input("Enter n value :"))

for i in range(90,90-n,-1):
    for j in range(90,90-n,-1):
        print(chr(i),end=' ')
    print()

```

```

Z Z Z Z Z
Y Y Y Y Y
X X X X X
W W W W W
V V V V V

```

In [2]: *# Pattern 21*

```

# *
# * *
# * * *
# * * * *
# * * * * *

n = int(input("Enter n value :"))
for i in range(1,n+1):
    for j in range(i):
        print("*",end=' ')
    print()

```

```

*
* *
* * *
* * * *
* * * * *

```

In [184...

```

# Pattern 22
# 1
# 1 2
# 1 2 3
# 1 2 3 4
# 1 2 3 4 5

n = int(input("Enter n value :"))

for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=' ')
    print()

```

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

In [12]: *# Pattern 23*

```

# A
# A B
# A B C
# A B C D

n = int(input("Enter n value :"))
for i in range(ord('A'),ord('A')+n+1):

```

```

for j in range(ord('A'),i):
    print(chr(j),end=' ')
print()

```

A
A B
A B C
A B C D

```

In [66]: # Pattern 24
# 1
# 2 3
# 4 5 6
# 7 8 9 10

n = int(input("Enter n value :"))
num = 1
for i in range(1,n+1):
    for j in range(i):
        print(num,end=' ')
        num +=1
    print()

```

1
2 3
4 5 6
7 8 9 10

```

In [67]: # Pattern 25
# A
# B C
# D E F
# G H I J

n = int(input("Enter n value :"))
num = 65
for i in range(1,n+1):
    for j in range(i):
        print(chr(num),end=' ')
        num +=1
    print()

```

A
B C
D E F
G H I J

```

In [69]: # Pattern 26
# * * * * *
# * * * *
# * * *
# * *
# *

n = int(input("Enter n value :"))
num = 65
for i in range(n):
    for j in range(n):
        print('*',end=' ')
    print()

```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [212... # Pattern 29
# 1
# 2 4
# 3 6 9
# 4 8 12 16
# 5 10 15 20 25
n = int(input("Enter n value :"))
for i in range(1,n+1):
    for j in range(1,i+1):
        print(i*j,end=' ')
    print()
```

```
1
2 4
3 6 9
4 8 12 16
5 10 15 20 25
```

```
In [214... # Pattern 30
# 5
# 5 4
# 5 4 3
# 5 4 3 2
# 5 4 3 2 1

n = int(input("Entre n value :"))

for i in range(n,0):
    for j in range(i,n-i-1):
        print(j,end='')
    print()
```

```
In [5]: ord('A')
```

```
Out[5]: 65
```

```
In [ ]:
```